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A comparison of brand equity strength across consumer segments and markets

Abstract

Purpose: This research compares *brand equity strength*, i.e. the extent to which brand awareness and brand image contribute to purchase propensity, for different segments of consumers (non-users, light users and heavy users) and two different markets (soft drinks and banking, representing a repertoire and a subscription context, respectively).

Approach: This aim is pursued using a scalable Customer Based Brand Equity (CBBE) framework, which captures how brand awareness and brand image, on a continuum of brand knowledge, underpin purchase propensity. The framework constitutes a ‘tool’ for the analysis of brand equity strength and it is applied, alongside a suite of empirical tests, to a large set of longitudinal consumer survey data collected from the same consumers and for both markets.

Findings: There are meaningful differences across the three consumer segments considered, especially in relation to brand image values, which are generally greater for more loyal consumers. Furthermore, the overall strength of brand equity is greater for banking brands compared to soft drinks brands.

Originality/value: The contribution of this research to brand equity knowledge is twofold. It addresses concerns in relation to the need to analyze brand equity at a disaggregated level; and it sheds light on inconclusive findings in relation to the generalizability of CBBE principles across different types of markets.

Practical implications: This research highlights the practical importance of detecting and managing differences in brand equity strength across consumer segments with dissimilar brand loyalty. It also suggests that there is relatively more value in evaluating and managing the CBBE process in subscription markets, than in repertoire markets.

Key words: *Customer Based Brand Equity, brand awareness, brand image, purchase propensity, brand choice, brand loyalty.*

1. Introduction

Keller's (1993) Customer Based Brand equity (CBBE) framework is one of the most widely used theories of brand equity. It is at the heart of current product and brand management research and practice, and it continues to attract efforts aimed at improving the understanding and measurement of brand equity (see Keller and Lehman, 2006). Out of these efforts, the majority has been directed at identifying and appraising the dimensions of brand equity *or* its outcomes (Christodoulides and de Chernatony, 2009). More recently, efforts have shifted towards linking these two aspects, highlighting the importance of determining the extent to which brand equity dimensions underpin its outcomes, leading to the conceptualization of CBBE as a *process* (Krystallis and Chrysochou, 2014; Christodoulides, Cadogan and Veloutsou, 2015; Chatzipanagiotou, Veloutsou and Christodoulides, 2016). Nonetheless, knowledge in the area progressed primarily through the use of complex models and/or deterministic approaches, obstructing the resolution of two issues, which represent the key aims of the present study.

First, existing research has not examined CBBE in sufficient detail at disaggregated level, i.e. for segments of consumers who differ in their levels of brand loyalty (Rust et al., 2004; and Christodoulides and de Chernatony, 2009). Works presenting disaggregated analyses have focused on individual dimensions of brand equity such as brand image, neglecting others (e.g. brand awareness) and have not considered any outcome of CBBE. They have also only compared particular consumer segments, overlooking others (e.g. Barnard and Ehrenberg, 1990; Oakenfull and McCarthy, 2010; Romaniuk and Nenycz-Thiel, 2013). This poses a threat to the use of CBBE frameworks from a product and brand management perspective, given that positive (or negative) interpretations might hide underlying differences across consumer segments and their level of brand loyalty.

Second, existing research does not offer clean-cut empirical evidence in relation to the generalizability of brand equity principles across different markets, especially goods vs. services (Krishnan and Hartline, 2001). Specifically, some researchers have argued that CBBE might be somewhat more important and/or different for services (e.g. Bharadaj et al., 1993; Onkovist and Shaw, 1989, Sharp, 1996), but have done so primarily through conceptual analyses (e.g. Berry, 2000), or without a direct comparison against goods (e.g. de Chernatony et al., 2004; Grace and O’Cass, 2005; Krystallis and Chrysochou, 2014). This is concerning, given that buying behavior literature highlights marked differences for a key outcome of brand equity, brand purchase propensity, for repertoire and subscription markets; a distinction that is conceptually akin to goods and services (see Sharp, Wright and Goodhardt, 2002).

This study addresses these two issues by using a simple and scalable framework, which theorizes CBBE as a process and *brand equity strength* as the outcome of that process, to capture the extent to which brand awareness and brand image underpin brand purchase propensity along a *continuum* of brand knowledge (see also Hoyer and Brown, 1990; Gordon et al., 1993; and Biedenbach and Marell, 2010). The framework is deployed as a ‘tool’ for the analysis of a large set of survey data ($N = 771$) gathered in the UK from the same consumers across multiple time periods and for two markets. In more detail, the present research appraises differences between three different segments of consumers (non-users, light users, and heavy users), and across a repertoire and a subscription market (soft drinks and banking). It does so using a suite of empirical tests derived from existing research (e.g. Agarwal and Rao, 1996; Kennedy and Ehrenberg, 2001; Maio Mackay, 2001; Punj and Hillyer, 2004), repeated across each time period to enhance the validity of the results and to reveal *objective* differences in the CBBE process and brand equity strength, without the need to resort to complex modeling techniques. Accordingly, this study addresses the following research questions: *Are there differences in the CBBE process across consumer segments, which are*

dissimilar in brand loyalty? Are there differences in the CBBE process between markets, which are dissimilar in brand purchase propensities?

Answering these questions contributes to theory and practice by clarifying two main aspects of CBBE. First, it elucidates that brand equity strength will depend upon the extent to which brand information retained in memory will evolve and strengthen over time along a continuum of brand knowledge, bolstering the chances of a brand to be chosen for purchase. Second, this study clarifies how this process might vary as result of inherent differences in brand loyalty across consumer segments and/or on the basis of market characteristics (e.g. repertoire vs. subscription). Combined, these insights could assist managers with better understanding of the objectives and impact of their marketing and branding initiatives.

2. Background

In origin, Keller (1993) described brand equity from a consumer perspective, defining Customer Based Brand Equity (CBBE) as a dynamic consumer response resulting from the retention (in memory) of *brand knowledge*, which he theorized into two specific dimensions: *brand awareness* and *brand image* (explained in detail in the next section). Specifically, he considered this dynamic response to be the overall strength of the brand manifested through fundamental differences in consumer behavior and differential consumer responses to specific marketing activities (Hoeffler and Keller, 2003).

The dynamic nature of Keller's (1993) CBBE has been embraced by others, such as Gordon et al. (1993), who conceptualized brand equity (albeit on the basis of limited empirical support) in terms of hierarchical effects between: i) CBBE dimensions in relation to each other; and ii) CBBE dimensions in relation to CBBE outcomes. Similarly, on the basis of the assumption that information about a brand is learned, retained and subsequently used to make choices, Erdem et al., (1999) theorized brand equity as the incremental effect that

consumer learning has on decision-making; this results in a probabilistic process, which establishes a *predisposition* to consider and evaluate (positively) a brand for purchase. These works contributed to shedding light on two key characteristics of CBBE, as follows. First, brand equity and its outcomes are linked via the cognitive mechanisms that come into play during the stages of consumer preferences and choice (see also Punj and Hillyer, 2004). Second, brand equity dimensions connect to each other *and* to CBBE outcomes in a ‘linear fashion’ (see also Esh et al., 2006).

More recently, these seminal ideas led to the conceptualization of *brand strength* as the evaluative and/or behavioral response resulting from strong configurable cognitive paths that are likely to affect brand choice (Grohs et al., 2016). Moreover, some of the latest contributions to CBBE measurement developed further assumptions on the dynamic and hierarchical nature of brand equity, highlighting the need to examine it as a *process* rather than a construct – e.g. Christodoulides et al., (2015); and Chatzipanagiotou et al., (2016); Krystallis and Chrysochou (2014). Specifically, a key underlying assumption common to these recent works is the following. The various dimensions of brand equity are expected to lead to *changes* in brand equity outcomes (see also Yoo and Donthu, 2001); and these changes, rather than the absolute value of the dimensions or the outcome variables considered, encompass CBBE (Christodoulides et al., 2015). Therefore, for a robust conceptualization and evaluation of brand equity, besides understanding and examining which dimensions make up brand equity, it is crucial to determine the extent to which each dimension impacts the outcomes (Christodoulides et al., 2015).

Although important, the research illustrated so far shows some inherent issues. Original works putting forward the idea of CBBE being dynamic and hierarchical are primarily based on complex models, which pose a threat to the generalizability of the research across markets and conditions, as well as restrict their practical use (Aaker, 1996). Early

works also show limited consensus on *how* the various cognitive dimensions of CBBE link to each other and evolve towards relevant outcomes, thus tangling the CBBE dimensions and outcomes. Most recent research examining CBBE as a process has partially addressed these flaws, but has often introduced hedonic and/or emotional dimensions of CBBE (e.g. Chatzipanagiotou et al., 2016); these ‘new’ dimensions are likely to depend upon a pre-existing connection with the brand, which limits applicability across different consumer segments (e.g. consumers at different stages of their ‘relationship’ with the brand). Moreover, from an empirical perspective, brand equity is easier to appraise and understand when focusing on the cognitive ‘chains of effects’ feeding into the various stages of consumer choice (Punj and Hillyer, 2004). Lastly, existing frameworks are primarily based on deterministic appraisals of CBBE (or the CBBE process, i.e. brand equity strength); yet, to fully capture how ongoing consumer learning underpins purchase decisions, it is preferable to consider CBBE outcomes as probabilistic events (Erdem et al., 1999).

2.1 Proposed conceptual framework

In line with the above discussion, the conceptual framework that this research proposes and uses for analytical purposes considers CBBE as a process resulting from a *continuum* of brand knowledge, featuring Keller’s (1993) key dimensions, i.e. brand awareness and brand image. Brand awareness is the simplest level of the continuum and the ‘starting point’ of the CBBE process, whereas brand image is a more developed psychological and cognitive structure, which enables the consumer to think about additional brand information (see Hoyer and Brown, 1990; Nedungadi, 1990; and Biedenbach and Marell, 2010). The outcome of this continuum is an overall probability to choose the brand, as represented by its *purchase propensity* (see also Grohs et al., 2016), which encompasses the overall *brand equity strength* (see also Gordon et al., 1993; Chatzipanagiotou et al., 2016). The use of a probabilistic approach (see also Erdem et al., 1999) maximizes the chances to quantify and understand the

extent to which the ongoing consumer learning process, along the continuum of brand awareness and brand image, enhances consumer predispositions to brand choice.

Additionally, the inclusion of a market performance dimension of CBBE in the framework enhances its practical relevance and generalizable use across different markets, in line with Aaker's (1996) recommendations.

Importantly, in line with Punj and Hillyer (2004), the framework assumes that brand loyalty and CBBE share similarities in relation to their attitudinal aspects, but differ greatly in their behavioral aspects. Hence, brand loyalty is considered 'exogenous' to the CBBE process. The following sections explain in greater detail each element of the framework.

2.1.1 Brand awareness

Brand awareness refers to a state of consumer knowledge about a brand (Hoyer and Brown, 1990) and the cognitive processes underpinning that knowledge, such as: i) perceiving the brand as being previously encountered (Mandler, 1980); ii) the ability to identify and recognize the brand (Percy and Rossiter, 1992; Hoyer and Brown, 1990; Aaker, 1991); or iii) simply 'knowing about the brand' (Huang and Sarigöllü, 2012). Keller (1993; 2003) provides a more advanced theorization of brand awareness, defining it as the combination of the ability to: i) recognize prior exposure to the brand (*brand recognition*) and ii) process the brand in memory whenever a stimulus is presented as a cue – e.g. the product category (*brand recall*).

Most research into brand awareness has either excluded it from the dimensions of CBBE (e.g. Lassar, Mittal and Sharma, 1995; Erdem et al., 1999; Bian and Mouthino, 2011) or simply combined it with other dimensions (Yoo, Donthu and Lee, 2000). A plausible reason for this is the fact that existing works tend to consider brand awareness as a rather rudimentary aspect of consumer knowledge, and have primarily focused on more complex aspects such as brand image (MacDonald and Sharp, 2000). Nonetheless, there are three key

reasons why this study includes brand awareness in the proposed framework and considers it a sub-component of brand knowledge on its own, likely to ‘start’ the CBBE process:

- i. According to Nedungadi and Hutchinson (1985), brand awareness is essential to *schema generation*, i.e. for knowledge abstraction from the product category to the brand. In fact, Keller (1993) refers to brand awareness as ‘category identification’.
- ii. Brand awareness can be a dominant *brand choice heuristic*, which helps consumers to reduce time and effort during decision-making (Bettman and Park, 1980; Hoyer and Brown, 1990). This is particularly relevant whenever consumers spend minimal effort in choosing brands, as per the instance of common repeat-purchase products (Hoyer, 1984). It is also relevant for complex purchase decisions to infer quality (Pappu et al. 2005) and to reduce risk (Bettman and Park, 1980; Hoyer and Brown, 1990).
- iii. Brand awareness represents a fundamental ‘learning advantage’ for the brand (Huang and Sarigöllü, 2012), which assists consumers during the stages of the decision-making process leading to choice (Nedungadi, 1990). Specifically, it underpins the chance that a consumer will include the brand in the consideration set and in their repertoire of purchases (Hoyer and Brown, 1990; Nedungadi, 1990; MacDonald and Sharp, 2000).

2.1.2 Brand image

As Stocchi et al. (2015) recently remarked, brand image combines cognitive as well as psychological aspects of how consumers perceive brands (see also Dobni and Zinkhan, 1990; and Patterson, 1999). This is in line with Keller’s (1993) original conceptualization of brand image, which is based on cognitive psychology and, in particular, on widely accepted models of human memory and information processing such as the Associative Network Theory (ANT) and Active Control of Thought (ACT-R) (see Anderson et al., 2004 for a recent review of these theories). Accordingly, Keller depicts brand image as the network of brand-related information that consumers retain in memory, whereby the brand represents the focal concept

and the information linked to the brand are classified as *brand image associations*. He further distinguishes brand associations as either explicating the brand as a concept or as embodying consumer evaluations. Namely, he distinguishes across: i) *attributes* or descriptive features of the brand (e.g. price, appearance, and imagery resulting from brand positioning); ii) perceived symbolic, experiential, and functional *benefits* (i.e. more or less intrinsic and hedonic features); and iii) *attitudes* or overall evaluations of the brand. Therefore, from a cognitive and psychological perspective, brand image is a more complex dimension of brand knowledge than brand awareness.

Although Keller has never operationalized brand image, his conceptualization has enabled other researchers to develop multi-item measures of brand equity and/or derive brand performance metrics. In relation to the second aspect, there is a stream of research that has built upon Keller's definition of brand image to assess the *mental availability* of brands in competitive settings (Romaniuk and Sharp, 2004; Romaniuk, 2013).

When assessing mental availability, Keller's conceptualization of brand image serves as a baseline framework to measure a brand's *mental market share*; that is, the ratio between a brand's total number of brand image associations and the total number of associations obtained by all brands in the product category (Romaniuk, 2013). Specifically, the associations used to calculate this measure can be derived from consumer surveys capturing perceptions with 'pick-any' questions (see Driesener and Romaniuk, 2006), to be designed in line with Keller's framework. This measure assesses the chance that a brand will 'come to mind' when consumers face purchase decisions, as a result of relying on the overall strength and width of the network of brand image associations retained in memory. As such, brand image is not only a more complex dimension of consumer knowledge; on a conceptual level, it is also arguably 'one step closer' to brand choice.

2.1.3 Purchase propensities

Existing research in consumer buying behavior defines purchase propensity as a brand's probability to be bought in the future, as resulting from the combination of two factors: i) how many consumers have purchased the brand in the past; and ii) how many times, on average, consumers have purchased the brand in the past (Wright, Sharp and Sharp, 2002).

Importantly, both aspects underpin a brand's *market share*, which can be derived from panel data or from consumer survey questions (Wright et al., 2002).

As a *probability measure*, purchase propensity (and thus market share) offers more accurate predictions of behavior (i.e. it offers a robust representation of the competition that brands face for choice) and a better discrimination between segments of consumers displaying different inherent intentions to buy, such as users vs. non-users (Wright and MacRae, 2007). Furthermore, previous research has demonstrated that purchase propensity is a dependent variable that is useful to any study concerned with detecting the effects of psychological variables on consumer choices (Wright and MacRae, 2007). In fact, as Wright et al., (2002) remark, probability measures are often confused with intentions; yet, "rather than being attitudinal constructs hypothesized to guide future behavior, they are [...] estimates of the chances of that future behavior" (p. 82). It is therefore not surprising to see that many brand equity models include purchase propensity and/or market share among their outcome variables (Agarwal and Rao, 1996; Erdem et al., 1999; Maio Mackay, 2001; Washburn and Plank, 2002; Punj and Hillyer, 2004; Esch et al., 2006; Bian and Moutinho, 2011).

In the context of process models of brand equity, purchase propensity (or market share) is particularly useful because it lends itself to capturing the extent to which brand awareness and brand image contribute to the consumer's likelihood to choose a brand (see also Gordon et al., 1993); it also captures a key market behavior CBBE dimension, i.e. it reflects the relative advantage of strong brands (Aaker, 1996). Interestingly, this assumption

is in line with other areas of marketing research such as word-of-mouth literature, whereby purchase propensity is deployed to draw inferences on the effectiveness of word-of-mouth by comparing its value before and after exposure to the referral (e.g. East et al., 2008; Nguyen and Romaniuk, 2013). Hence, this research assumes that purchase propensity (and thus market share) will reflect, to a high degree of accuracy, any variation in the general likelihood of brand choice resulting from the continuum of brand awareness and brand image, encompassing the overall strength of brand equity (see also Srinivasan et al., 2010).

3. Research hypotheses

3.1 CBBE at the disaggregated level

There is a long-standing body of empirical research (Bird et al., 1970; Castleberry and Ehrenberg, 1990; Barnard and Ehrenberg, 1990) demonstrating that brand usage, as claimed in the context of consumer surveys, underpins how consumers perceive brands and purchase probability. A key outcome of this body of research has been the identification of concrete differences in brand image between consumer segments varying in frequency and/or recency of brand use and thus their level of *brand loyalty*. For example, early research demonstrated that the proportion of people associating a brand with a given attribute in the context of consumer surveys is generally: i) higher for current users compared to former users, and ii) very low for those who have never used the brand (Bird et al., 1970). Subsequent research has showed that the percentage of a brand's users who will associate the brand with given attributes generally increases with their claimed frequency or recency of purchasing, i.e. in line with whether consumers classify as light or heavy users of the brand (Barwise and Ehrenberg, 1985; Barnard and Ehrenberg, 1990).

More recently, Oakenfull and McCarthy (2010) compared the number and strength of brand image associations for light and heavy users across multiple product categories using

consumer survey data. These authors found that, in comparison to light users, heavy users retain in memory more and stronger brand image associations; they also retain more and stronger unique brand image associations, i.e. associations that are specific to that one brand only. Similarly, Romaniuk and Nenycz-Thiel (2013) analyzed data from two packaged goods categories and found that 100% loyal customers (those exclusively loyal to only one brand in a given time period) give more brand image associations than any other customers.

Arguably, a key underlying limitation of these works is that they focus only on brand image, with no explicit consideration for the other components of CBBE or the overall brand equity strength. Moreover, the above research only compared two sub-samples of consumers at a time (e.g. users vs. non-users; or light vs. heavy users). Accordingly, the present research expands upon these works by simultaneously comparing three consumer segments (non-users, light users and heavy users), while focusing on multiple components of CBBE (brand awareness and brand image) and considering, more specifically, the overall strength of brand equity (purchase propensity). Specifically, on the basis of the empirical evidence discussed so far and in line with the conceptual framework that this research uses, it is plausible to: i) extend existing findings from brand image to the whole continuum of brand knowledge; and ii) combine all findings available (i.e. for non-users vs. users and light users vs. heavy-users) into the following research hypotheses:

H1: *The strength of brand equity will be greater for brand users than for non-users.*

H2: *The strength of brand equity will be greater for heavy users than for light users.*

3.2 CBBE generalizability across markets

Consumer buying behavior literature highlights some marked differences in brand purchase probability between *repertoire* and *subscription* markets, resulting from clear patterns in repeat purchases and brand loyalty (Sharp, Wright and Goodhardt, 2002). In repertoire markets, consumers tend to minimize efforts in purchase decisions and routinely purchase out

of small repertoires of preferred brands (Sharp et al., 2002). This typically leads to low levels of brand loyalty and only minimal differences in the frequency of purchases, as shown in Fast Moving Consumer Goods (FMCGs) markets such as soft drinks, laundry detergents, and cereals. Conversely, in subscription markets, consumers primarily purchase one main brand and exhibit high levels of brand loyalty as well as much smaller brand repertoires (Zeithaml, 1981; Sharp et al., 2002). Examples of subscription markets include services categories such as banking and insurance. Importantly, these differences across markets are not due to differences in inter-purchase time, but rather, they reflect inherent differences in brand purchase propensities (Sharp et al., 2002).

Although clearly relevant to a key CBBE outcome, purchase propensity, CBBE research does not commonly refer to the distinction between repertoire and subscription markets. Nonetheless, existing works that have challenged Keller's (1993) and/or Aaker's (1991) claim of the generalizability of CBBE principles across markets, inadvertently corroborate the empirical differences in purchase propensity between repertoire and subscription markets. For example, Berry, (2000), de Chernatony et al., (2004), Grace and O'Cass, (2005) and Krystallis and Chrysochou, (2014) presented service-specific CBBE frameworks and revealed that, although brand equity for service brands also originates from brand awareness and brand image, there are some differences for services relative to goods in terms of: i) the importance of each dimension of brand equity, and ii) the type of information and/or experiences that consumers retain in memory. In relation to the first point, some scholars have argued that brand image plays a greater role in CBBE for services brands than for goods (e.g. Berry, 2000). In relation to the second point, these works identified that the experiential nature of the service, the impact of the physical environment where the service delivery occurs and the personal interaction with front-line employees make the image of services brands more important than goods (O'Cass and Grace, 2003; Grace and O'Cass,

2005; Krysrallis and Chrysochou, 2014). Accordingly, some authors contended that brand equity is more important for services brands (e.g. Onkovist and Shaw, 1989; Bharadwaj et al., 1993). Nonetheless, this literature is based on limited empirical evidence and has not examined the dimensions and outcomes of CBBE together (or the overall brand equity strength).

In addition to the above, it is worthwhile considering the following aspects. Firstly, due to the inherent difficulty in evaluating service brands prior to choice (Murray, 1991), brands play an especially important role for services by enabling a pre-assessment of service quality (Bharadwaj et al., 1993; Murray, 1991; Zeithaml, 1981; Davis, 2007) and providing a means to differentiate across alternatives (de Chernatony and Dall’Olmo Riley, 1999; McDonald et al., 2001). Secondly, due to the intangible nature of services, brands provide a mechanism for consumers to ‘tangibilize the intangible’ (Berry, 2000; Laroche et al., 2004), and enable the consumer to mentally and/or physically ‘grasp’ services (Laroche, Bergeron and Goutland, 2001). This helps to reduce perceived risk (Murray and Schlater, 1990; Laroche et al., 2004), to minimize uncertainty (Rathmall, 1966) and to build brand trust (Javalgi et al., 2006), all of which are considered inherent to service purchases. Consequently, due to the intangible nature and the high risk associated with services, consumer perceptions of services brands are often assumed to be particularly crucial to any behavioral response towards the brand (Davis et al., 2000).

Taken together, all aspects mentioned so far suggest that in the context of the overall CBBE process, it is reasonable to expect greater reliance on the continuum of brand knowledge during the steps leading to brand choice for services brands. If supported empirically, this mechanism could explain differences in purchase propensity between subscription and repertoire markets and prove claims of greater importance of CBBE for service brands. Accordingly, the present research tests following hypothesis:

H3: *The strength of brand equity will be greater for subscription markets than for repertoire markets.*

The following sections present the research methods used to test these hypotheses, together with an overview of the empirical results and relative implications for theory and practice.

4. Methods

4.1 Data

This research is based on the analysis of a longitudinal UK consumer survey ($N = 771$), which featured brand awareness, brand image and stated purchase behavior data for a repertoire market, *soft drinks* (featuring data for 22 brands) and a subscription market *banking* (featuring data for 14 brands). A commercial market research company (Crosstabs) administered the survey and managed the sample of respondents over three consecutive waves of data collection in 2012, three weeks apart. Importantly, the same consumers were surveyed across all waves of data collection and for both markets, adopting a repeated measures design. Collectively, these characteristics mimic controlled experimental conditions, which allow for robust, objective testing of differences in the strength of brand equity by consumer segment and market. The sample included 51% females and 49% males; 6% of respondents were between 18 and 24 years, 33% between 25-44 years, 37% between 45-64 years, and 25% 65 years and above.

This study uses the information on current purchase behavior available in the data to identify the three user-group segments of interest: *non-users*, *light users*, and *heavy users*. Specifically, the user groups were identified for each individual brand, as follows.

For soft drinks, relying on the cut-off value of frequency and recency of purchase that previous works (e.g. Ehrenberg et al., 2004) highlighted, this research classifies: i) respondents who stated that they did not purchase the brand in the four weeks preceding the

survey as *non-users*; ii) respondents who claimed that they bought the brand once or twice in the past four weeks as *light users*; and iii) respondents who indicated that they purchased the brand more than two times in the past four weeks as *heavy users*.

For banks, in line with the concept of *first brand loyalty*, i.e. the tendency to satisfy most category needs with one single provider (East et al., 1995), this research classifies: i) respondents who stated that they were not a current customer of the bank at the time of the survey as *non-users*; ii) respondents who claimed that they were current customers of the bank though it was not their main financial institution as *light users*; and iii) respondents who indicated that they were current customers of the bank and that it was their main financial institution as *heavy users*.

The above classifications were repeated, independently, for each brand available in the two markets, and across all time periods. Appendix A features all sub-samples sizes per segment and per each brand, by time period and in both markets. Last, to control for the uneven sizes of the segments identified, this research bases all empirical tests on brand-level measures, averaged and compared *between* sub-samples, and described next.

4.2 Measures

In line with the conceptual framework described before, the key measures that this research examines are (see also Appendix A):

- **Brand awareness**, measured as the percentage of *top of mind* brand awareness, i.e. the percentage of respondents who mentioned the brand first when prompted with the category (Nedungadi and Hutchinson, 1985).
- **Brand image**, measured as the percentage of *mental market share*, i.e. the percentage of brand image associations elicited for a brand out of the total brand image associations elicited for the category (Romaniuk, 2013). This study also considers two sub-components of this measure; i.e. *associative penetration* and *associative rate*. Associative penetration

is the proportion of consumers who elicited at least one brand image association for a brand. Associative rate refers to the average number of brand image associations that a consumer elicited for a given brand. Importantly, to ensure an un-biased and empirically sound measurement, all three measures were derived using Romaniuk's (2013) multi-steps approach to modeling pick-any brand image associations.

- **Purchase propensity**, measured as *market share*, in line with Wright et al., (2002)'s rationale. Specifically, based, in part, on the same data used to classify the user segments, purchase propensity was derived for each individual brand in the data set from: i) the proportion of respondents stating that they purchased a certain brand in the past four weeks vs. all other brands for soft drinks; and ii) the proportion of respondents stating that they were currently customers of a certain bank vs. all others banks for banking.

4.3 Empirical tests

To test H1, H2 and H3, this study deploys a suite of empirical tests based on validated approaches from previous research, as follows.

First, it uses Mean Absolute Deviations (MAD) (Kennedy and Ehrenberg, 2001) to compare the absolute differences in the values of the measures across the consumer segments and all time periods. This is an essential condition for testing and validating H1 and H2, given that if measures do not display differences, it is not possible to test differences in the strength of brand equity between the segments considered. Specifically, this method is based on the comparison of the values of brand awareness and brand image measures at the individual brand level between the following segments: *all users vs. non-users*; *all users vs. light users*; *all users vs. heavy users*; *non-users vs. light users*; *non-users vs. heavy users*; *light users vs. heavy users*. In line with existing works, this research treats MAD values equal to, or greater than 5%, as an indication of a substantial difference in the values of the measures between segments (Kennedy and Ehrenberg, 2001). For ease of interpretation and to further compare

between the two markets being considered, the analysis focuses on the averaged MAD scores across all brands and across the three time periods.

Second, to test more explicitly H1, H2 and H3, this study examines the convergence of measures through pair-wise Pearson's correlations (Agarwal and Rao, 1996; Maio Mackay, 2001) and a multiple regression analysis (Punj and Hillyer, 2004). Both methods were deployed to reveal, for the different segments and the two markets considered, the extent to which brand awareness, mental market share, associative penetration, and association rate explain variance in purchase propensity.

The use of multiple empirical tests repeated across several time periods enhances the validity of the results and reveals *objective* differences in the strength of brand equity across consumer segments and markets, without the need to resort to complex modeling techniques.

5. Results

5.1 Differences in the values of the measures

For the soft drinks market (see Table 1), the comparison of the values of all measures considered revealed some clear patterns, which were consistent across all time periods. In terms of brand awareness, there were differences between non-users and heavy users, and light users and heavy users (MAD was on average 7% and 6%, respectively). There were, however, more differences in the values of the associative penetration and the associative rate. Specifically, for associative penetration, differences ranged between 29% and 34% when comparing the whole sample against light and heavy users; and up to 37% when comparing non-users and heavy users. Differences were even more pronounced for associative rate, with an average MAD score of 55% between the whole sample and light users, and 110% between the whole sample and heavy users. Similarly, there was more than 70% difference between

non-users and light users as well as between light users and heavy users. For non-users and heavy users the difference was over 127%.

*** Insert Table 1 about here ***

Taken together, these results reveal the existence of large differences across segments in brand awareness, albeit primarily among the most loyal consumers. They also reveal more substantial differences across segments for brand image, especially for the associative rate, (light and heavy users are able to provide far more associations than any other segment).

When considering banking (see Table 2), there were identical patterns in the differences across segments, consistent across all time periods. Moreover, many of the differences were somewhat greater for this market in comparison to soft drinks (see also Table 3 for a direct comparison of the two markets). In more detail, the average MAD for brand awareness between the whole sample and heavy users was 21%, a value underpinned by a difference of 24% between non-users and heavy users, and a 20% between light users and heavy users. With regard to associative penetration, the average MAD scores ranged between 28% (whole sample vs. light users) and 45% (non-users vs. heavy users). Finally, the associative rate displayed the largest differences of all, with average MAD scores always greater than 100% (182% between non-users and heavy users; 150% between the whole sample and heavy users; and 129% between light and heavy users).

*** Insert Table 2 about here ***

*** Insert Table 3 about here ***

These results are an essential condition to testing the validity of H1 and H2, i.e. to verify that the strength of brand equity is greater for users vs. non-users (H1), and greater for heavy users vs. light users (H2). Furthermore, these results reveal that there are greater differences between segments and across markets in relation to brand image, rather than brand awareness; i.e. differences may manifest themselves at a more advanced level of the continuum of consumer knowledge. They also suggest that differences are generally more marked in banking (H3).

5.2 Outcome of hypotheses testing

The convergence analysis for the category of soft drinks (see Table 4) showed that for the whole sample, all measures were generally significantly correlated ($p < 0.01$ in 90% of the instances, two-tailed) across all time periods, and correlations were also always greater than 0.61, which would be considered strong (as per Cohen, 1988). However, there were some underlying differences across segments, in line with H1 and H2.

For non-users of soft drinks there were hardly any significant correlations, the exception being for brand awareness and the associative rate (0.65 on average across all three time periods, always significant at the 0.01 level). This confirms that consumers who are currently not using a certain brand are generally aware of it, which enables them to elicit at least some brand-image associations; however, this is not strong enough to underpin purchase propensities. In contrast, light and heavy users of soft drinks showed two clear patterns, as follows. There was greater convergence between brand awareness and mental market share, and between these two measures and purchase propensities. Interestingly, these correlations with purchase propensity yielded similar values for light and heavy users (i.e. 0.69 and 0.65 for brand awareness; and 0.92 and 0.96 for mental market share, in both instances significant at the 0.01 level); notably with greater convergence between mental market share and purchase propensity.

*** Insert Table 4 about here ***

In the banking market (Table 5), all measures were significantly correlated ($p < 0.01$, two-tailed) across all time periods with no exceptions, and they were always greater than 0.76. This outcome suggests that there is greater overall convergence of the CBBE measures considered for the banking market than for soft drinks, this being further indication that there is support for H3.

*** Insert Table 5 about here ***

The detailed analysis of correlations for the different segments of consumers revealed additional patterns that corroborate H1, H2 and H3. First, for non-users of banking brands, there were significant correlations between mental market share and associative penetration (0.94, $p < 0.01$); between mental market share and associative rate (0.90, $p < 0.01$); and between associative penetration and associative rate (0.84, $p < 0.01$). Second, for light users and heavy users, correlations for purchase propensity were especially large and significant. For example, for light users there was a correlation of 0.92 (significant at the 0.01 level) between mental market share and purchase propensity; and for heavy users there was a correlation of 0.92 between brand awareness and purchase propensity, and 0.98 between mental market share and purchase propensity (both significant at the 0.01 level).

To explore and explain these outcomes in greater depth, this research finally considers the results of a multiple regression analysis, using purchase propensity (expressed as market share) as the outcome (dependent) variable. As Table 6 and 7 show, in both markets and for the whole sample, all four measures combined significantly predicted variance in purchase

propensity ($p < 0.001$; R^2 ranging between 0.75 and 0.95 across time periods and all measures); yet, no measure individually predicted purchase propensity. Furthermore, the measures generally predicted more variance in purchase propensity in the banking than soft drinks (as depicted by higher R^2 values).

*** Insert Table 6 about here ***

*** Insert Table 7 about here ***

The same finding was consistent when considering the segments of light and heavy users. In both markets, all models were again significant ($p < 0.001$) across all time periods and greater than for the aggregate user level. Moreover, the measures explained, on average, slightly more variance in purchase propensity for the banking market than for the soft drink market (average R^2 was 0.96 for banking and 0.94 for soft drinks). Finally, mental market share consistently (i.e. for both markets and all time waves) predicted variance in purchase propensity with a high degree of accuracy ($p < 0.001$) for both light and heavy users.

Taken together, the outcomes of the convergence and regression analyses further validate H1, H2 and H3. Hence, results suggest that the strength of brand equity varies across different consumer segments and product categories and is greater for heavy users and in subscription markets.

6. Discussion

This research evaluated differences in the strength of brand equity for different consumer segments and different markets. Specifically, this study compared brand equity strength for consumer segments that differed in their levels of brand loyalty, and for two markets, which

are inherently dissimilar with respect of the typical levels of brand purchase propensity that consumers display.

The results highlighted some key overarching patterns consistent across multiple time periods, which yield theoretical and practical implications, as follows.

6.1 Theoretical implications

The analysis of brand equity strength (and thus the CBBE process) at the disaggregate level highlighted that there are differences between the *most loyal* consumers (heavy users) and other segments in subscription markets, and between the *least loyal* consumers (light users) and other segments in repertoire markets; moreover, these differences are more marked for subscription markets. These findings add to existing knowledge in several ways, as follows.

Firstly, the majority of existing works in brand equity research are based on aggregate analyses. Works that have considered specific segments focused only on brand image and did not include other dimensions and/or outcomes of CBBE; they also compared only some segments at the time (e.g. users vs. non-users *or* light vs. heavy users). In contrast, this study offers a comprehensive examination of CBBE at the disaggregate level, and shows that that there are some differences worthwhile considering across segments along the whole continuum of brand knowledge. Such differences underpin brand propensity and thus the overall brand equity strength resulting from the CBBE process. Accordingly, it is recommendable to establish and compare segment-level brand equity appraisals at least in relation to brand image, to reveal and explain deficits or advantages in brand equity strength, which aggregation would otherwise hide or contaminate.

Secondly, this study contributes to shedding light on the inconclusive findings of research that has debated the generalizability of brand equity principles across markets, and validates assumptions in relation to the fact that brand equity is at least as important for services as it is for goods (e.g. Berry, 2000; Onkovist and Shaw, 1989; Bharadwaj et al.,

1993). Specifically, thanks to the repeated comparison (across multiple time periods) of data by the same consumers for two markets (soft drinks and banking, a repertoire and subscription market), this study shows that consumers draw upon the whole continuum of brand knowledge when choosing service brands, and they do so to a greater extent than they do for goods. This outcome corroborates and extends empirically: i) research theorizing that consumers use brand awareness as a choice heuristic to infer product quality for service brands (Pappu et al., 2005) and to reduce purchase risk (Bettman and Park, 1980; Hoyer and Brown, 1990); and ii) Berry's (2000) assumption in relation to the importance of building and reinforcing brand image for services, given that consumers will often draw upon more complex brand information in an attempt to further reduce risk (see also Murray and Schlater, 1990; and Laroche et al., 2004). Accordingly, this research addresses an inherent bias towards CBBE research in goods markets (Christodoulides and de Chernatony, 2009); and provides much needed insight into CBBE in service markets and how it compares to goods markets; which is important given the growth of service brands (O'Cass and Grace, 2005).

Finally, the identification of marked differences across markets adds significantly to buying behavior literature. That is, in line with existing buying behavior research (e.g. Sharp et al., 2002; Sharp et al., 2012), the outcomes of this study support the existence of differences between repertoire and subscription markets. At the same time, the findings also expand knowledge in the area, by shedding light on the underlying reasons behind the differences in purchase behaviors in each market; i.e. differences in the process according to which the continuum of brand knowledge underpins purchase propensities. This is an outcome that previous research has not acknowledged, and that has relevance and implications for the advancement of buying behavior research, especially in relation to its link to brand equity.

6.2 Practical implications

Brand equity appraisals provide particularly rich insights to assist researchers and managers with an understanding of the driving forces behind current purchase behavior (Christodoulides and de Chernatony, 2009; Srivastava and Shocker, 1991). The present study is no exception and offers two key practical implications of managerial relevance, as follows.

First and foremost, the results of this research indicate the importance of exploring CBBE as a process, by looking at how its key dimensions ‘evolve’ in a continuum of brand knowledge and underpin brand purchase propensity. This approach allows extracting more in-depth insights from consumer survey data; specifically, it offers a simple way to appraise the overall brand equity strength without having to resort to complex techniques, yet capturing a theoretically robust reflection of the CBBE process.

Furthermore, this study clearly suggests that disaggregate level brand equity appraisals are useful to accurately interpret brand performance, ultimately leading to the improvement of product and brand management strategies. In particular, the results of this research highlight the potential information that can be extracted by identifying different consumer segments on the basis of brand loyalty and evaluating their level of brand awareness and brand image, in respect of brand purchase propensity. Accordingly, managers could come up with tailored actions aimed at: i) building from scratch the brand knowledge (starting with brand awareness) of consumers that show limited or no brand loyalty, e.g. via using wide-reach advertising and branding campaigns; ii) enhancing the brand knowledge (along the continuum triggered by brand awareness) of consumer who show limited brand loyalty, e.g. using line extensions to reinforce brand image; and iii) ensuring that the continuum of brand knowledge of consumers who are highly loyal successfully translates in brand buying, e.g. by enhancing availability in stores and visibility on shelf.

Lastly, the findings of this study indicate that managers should specify unique expectations in relation to the overall brand equity process and resulting brand management implications according to the market in which they are operating. Specifically, although the underlying mechanism according to which brand awareness and brand image underpin brand purchase propensities is essentially the same, there are some differences that must be taken into account according to whether a brand operates in a repertoire (goods) or subscription (services) market. Above all, it is useful for managers to know that consumers rely on a continuum of brand knowledge (especially brand image) to a greater extent in subscription markets to reduce risk and simplify brand purchase decisions. Therefore, gathering and monitoring information in relation to the brand equity process is particularly crucial to understanding CBBE for subscription markets and, more generally, for services.

7. Conclusions, limitations and future research

In summary, the present research has offered a detailed analysis for different consumer segments and two markets, shedding light on a series of novel findings in relation to how brand equity strength differs. This outcome has several theoretical and practical implications, which highlight the need to consider brand equity at the disaggregate level, and appraise brand performance in relation to brand equity differently according to the market.

Nonetheless, this work has some limitations, which could be addressed with future research. First, this study is based on the analysis of a single data set collected in one country. Thus, although the data allowed controlling for variation across time and between samples, it is desirable to replicate the same analysis across different contexts; for instance, by expanding across other product categories and other countries. Second, although scalable, the conceptual framework used for the analysis of CBBE is rather simplistic, focusing on consumer knowledge. Hence, future research could deploy an expanded version of this framework, and

in addition to examining purchase propensity it could include other variables such as price premium. Third, the empirical tests used were selected to answer specific research questions; however, future research could consider investigating further differences between consumer segments and markets using a structural equation modeling approach to uncover antecedents and moderating factors. Finally, to cater for the high degree of convergence of the brand awareness and brand image that the results showed, future works could examine these measures by focusing on the cognitive processes that underpin them, such as information retrieval (Alba and Chattopadhyay, 1985; Nedungadi, 1990; Holden and Lutz, 1992; 1993; Romaniuk and Sharp, 2004) and relative sub-components of it (e.g. familiarity and biographical recollection, see Stocchi et al., 2016). However, a possible drawback of such an approach would be conflating brand awareness and brand image measures, and thus possibly limiting the possibility to draw independent conclusions on the implications of each to marketing and branding strategies. Nonetheless, as Christodoulides et al., (2015) recently argue, given the theoretical ‘closeness’ of brand awareness and brand image, there is no guarantee that it is possible to completely overcome this issue.

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Appendix A

Sub-samples' sizes for different segments of consumers

<i>Soft drinks:</i>	Non-users			Light users			Heavy users		
	W1	W2	W3	W1	W2	W3	W1	W2	W3
7-Up	660	669	698	32	73	60	21	25	13
Coca Cola	500	540	572	63	108	119	105	102	80
Coke Zero	681	702	708	21	42	38	33	33	25
Diet Coke	574	611	617	39	69	81	82	76	73
Diet Pepsi	660	685	709	35	52	34	32	22	28
Dr Pepper	694	706	712	18	43	45	23	14	14
Fanta	629	641	675	27	82	69	45	41	27
Irn Bru	692	706	720	17	41	34	21	21	17
Lilt	698	718	741	18	48	23	7	4	7
Pepsi	640	667	665	29	66	69	41	43	37
Pepsi Max	632	669	672	35	55	50	51	45	49
Sainsbury	737	734	721	10	19	28	12	13	22
Schweppes	678	703	719	28	49	39	23	17	13
Sprite	649	679	696	32	80	59	28	24	16
Tango	661	704	722	31	55	34	23	16	15
Tesco	703	724	702	15	21	36	20	16	33
Ting	764	766	767	4	2	3	2	3	1
Powerade	714	719	741	14	29	20	11	16	10
Vitamin Water	736	746	746	12	12	14	11	14	11
Red Bull	688	697	724	12	36	18	39	36	29
Lucozade	615	648	657	31	65	60	59	66	54
Tesco Everyday value	737	743	731	4	16	19	12	12	21

W= wave of data collection

Note: In time period 1 and 2, the totals of the non-users, light users and heavy users for each brand do not total to 771 (the sample size). This is because of some missing values in the data.

<i>Banking:</i>	Non-users			Light users			Heavy users		
	W1	W2	W3	W1	W2	W3	W1	W2	W3
Barclays	610	598	594	62	75	77	99	98	100
C+G	756	748	746	15	23	24	0	0	1
Clydesdale	761	753	749	5	13	17	5	5	5
First Direct	745	728	727	9	22	25	17	21	19
Halifax	579	572	578	104	106	110	88	93	83
HSBC	665	651	648	27	45	44	79	75	79
Lloyds TSB	593	586	578	46	60	61	132	125	132
Natwest	615	609	608	48	58	59	108	104	104
Northern Rock	737	730	731	31	36	39	3	5	1
RBS	707	694	699	22	34	32	42	43	40
Sainsbury Bank	733	721	722	38	47	49	0	3	0
Santander	594	595	586	87	91	101	90	85	84
Standard Chartered	765	765	758	5	6	12	1	0	1
Tesco Bank	704	691	689	65	78	78	2	2	4

W= wave of data collection
C+G = Cheltenham and Gloucester
RBS = Royal Bank of Scotland

Details of measures (survey questions)

Measures	Survey questions	
	<i>Soft drinks</i>	<i>Banks</i>
Top of mind brand awareness (adapted from Nedungadi and Hutchinson, 1985)	<i>Which brands of soft drinks can you name?</i>	<i>Which companies that could offer you banking services can you name?</i>
Brand image (question based on Dreisener and Romaniuk, 2006; measurement based on Romaniuk, 2013)	<i>You will now see some qualities that people have linked to soft drink brands / banks. Can you look at each quality and indicate which, if any of the listed brands, you link to that quality? You can name as many or as few brands as you like, it doesn't matter if you have tried the brand or not.</i> <i>Note: this is a 'pick-any' question to elicit brand image associations as they come to mind to respondents, i.e. respondents are prompted with all of the brands in the market and a list of brand attributes (selected by marketing professionals) and can provide as many brand-to-attributes associations as they wish.</i>	
Purchase propensity (transformed into <i>market share</i> , based on the rationale of Wright, Sharp and Sharp, 2002)	<i>Which of the following soft drinks brands have you bought, either for yourself or others, in the past 4 weeks?</i>	<i>Which of the following companies are you currently a customer of?</i>

Biographies:

Lara Stocchi

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Dr Stocchi is Senior Lecturer in Marketing at Flinders University in South Australia, following roles at Loughborough University in the UK, Università Carlo Cattaneo LIUC in Italy and the University of South Australia in Adelaide. Dr Stocchi is developing an international reputation as a thought leader developing new understandings of consumer behavior and consumer memory. Working in collaboration with a wide range of industry partners in both the Fast Moving Consumer Goods and services sectors, she has analysed large datasets to develop new models to explain purchasing patterns. She has also developed new models of memory association towards products and brands, to capture the thoughts and feelings of consumers. Other areas of research interest include digital marketing (consumer perceptions of mobile applications), branding and the measurement of brand performance, and advertising effectiveness.

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Rachel Fuller is a doctoral researcher at the Centre for Service Management at Loughborough University School of Business and Economics. Her research interests include brand equity, consumer memory and decision-making, and her experience so far has focused on examining how these differ across markets/purchase contexts. Her work has been presented at the Frontiers in Service Conference and the European Marketing Academy (EMAC). Rachel is an experienced tutor, having taught undergraduate and postgraduate modules at both Loughborough University and Flinders Business School, at Flinders University in South Australia.

Tables and figures

Table 1 – MAD scores for soft drinks (all time periods, average of all individual brands)

		Non-users vs. whole sample	Light users vs. whole sample	Heavy users vs. whole sample	Non-users vs. light users	Non-users vs. heavy users	Light users vs. heavy users
Brand awareness	Time 1	0.01	0.05	0.06	0.06	0.07	0.05
	Time 2	0.01	0.02	0.05	0.02	0.05	0.04
	Time 3	0.01	0.03	0.10	0.04	0.10	0.08
	<i>Average</i>	0.01	0.03	0.07	0.04	0.07	0.06
Mental market share	Time 1	0.01	0.01	0.02	0.02	0.03	0.02
	Time 2	0.01	0.01	0.02	0.02	0.03	0.02
	Time 3	0.01	0.02	0.02	0.02	0.03	0.01
	<i>Average</i>	0.01	0.01	0.02	0.02	0.03	0.02
Associative penetration	Time 1	0.05	0.28	0.34	0.33	0.39	0.11
	Time 2	0.03	0.24	0.31	0.28	0.34	0.12
	Time 3	0.03	0.35	0.36	0.38	0.39	0.09
	<i>Average</i>	0.04	0.29	0.34	0.33	0.37	0.10
Associative rate	Time 1	0.19	0.58	1.07	0.77	1.27	0.67
	Time 2	0.15	0.58	1.13	0.73	1.29	0.80
	Time 3	0.15	0.50	1.09	0.64	1.24	0.67
	<i>Average</i>	0.16	0.55	1.10	0.72	1.27	0.71

MAD scores $\geq 5\%$ **bolded**

Table 2 – MAD scores for banking (all time periods, average of all individual brands)

		Non-users vs. whole sample	Light users vs. whole sample	Heavy users vs. whole sample	Non-users vs. light users	Non-users vs. heavy users	Light users vs. heavy users
Brand awareness	Time 1	0.03	0.04	0.19	0.06	0.22	0.16
	Time 2	0.03	0.02	0.25	0.04	0.28	0.24
	Time 3	0.03	0.02	0.20	0.04	0.23	0.20
	<i>Average</i>	0.03	0.03	0.21	0.05	0.24	0.20
Mental market share	Time 1	0.02	0.03	0.04	0.04	0.05	0.05
	Time 2	0.02	0.02	0.04	0.03	0.05	0.04
	Time 3	0.02	0.03	0.03	0.03	0.05	0.05
	<i>Average</i>	0.02	0.03	0.03	0.04	0.05	0.05
Associative penetration	Time 1	0.05	0.26	0.36	0.31	0.40	0.22
	Time 2	0.04	0.40	0.39	0.44	0.44	0.15
	Time 3	0.04	0.18	0.47	0.23	0.52	0.31
	<i>Average</i>	0.04	0.28	0.40	0.32	0.45	0.23
Associative rate	Time 1	0.28	0.71	1.45	0.95	1.77	0.97
	Time 2	0.26	0.49	1.46	0.55	1.76	1.35
	Time 3	0.27	0.27	1.59	0.39	1.95	1.56
	<i>Average</i>	0.27	0.49	1.50	0.63	1.82	1.29

MAD scores $\geq 5\%$ **bolded**

Table 3 – Comparison of MAD scores of soft drinks vs. banking (average of all time periods)

	Non-users vs. whole sample		Light users vs. whole sample		Heavy users vs. whole sample		Non-users vs. light users		Non-users vs. heavy users		Light users vs. heavy users	
	SD	B	SD	B	SD	B	SD	B	SD	B	SD	B
Brand awareness	0.01	0.03	0.03	0.03	0.07	<u>0.21</u>	0.04	0.05	0.07	<u>0.24</u>	0.06	<u>0.20</u>
Mental market share	0.01	0.02	0.01	0.03	0.02	0.03	0.02	0.04	0.03	0.05	0.02	0.05
Associative penetration	0.04	0.04	<u>0.29</u>	0.28	0.34	<u>0.40</u>	<u>0.33</u>	0.32	0.37	<u>0.45</u>	0.10	<u>0.23</u>
Associative rate	0.16	<u>0.27</u>	<u>0.55</u>	0.49	1.10	<u>1.50</u>	<u>0.73</u>	0.63	1.27	<u>1.82</u>	0.70	<u>1.29</u>

SD = soft drinks; B = banking.

MAD scores $\geq 5\%$ **bolded**; greatest MAD score between the two markets underlined

Table 4 – Convergence analysis for soft drinks (average of all time periods per segment)

	Brand awareness	Mental market share		Associative penetration		Associative rate		Purchase propensity	
Brand awareness	-	All	0.60**	All	0.43	All	0.67**	All	0.70**
		NU	0.33	NU	0.36	NU	0.65**	NU	-
		LU	0.80**	LU	0.32	LU	0.42	LU	0.69**
		HU	0.52**	HU	0.33	HU	0.05	HU	0.65**
Mental market share	-			All	0.95**	All	0.86**	All	0.75**
				NU	0.96	NU	0.73	NU	-
				LU	0.48**	LU	0.68**	LU	0.92**
				HU	0.45*	HU	0.31	HU	0.96**
Associative penetration	-					All	0.72**	All	0.60**
						NU	0.67	NU	-
						LU	0.45*	LU	0.33
						HU	0.09	HU	0.38
Associative rate	-							All	0.85**
								NU	-
								LU	0.54*
								HU	0.18
Purchase propensity								-	

All = whole sample; NU = non-users; LU = light users; HU = heavy users

** = correlation is significant at the 0.01 (two-tailed); * = correlation is significant at the 0.05 level (two-tailed)

Table 5 – Convergence analysis for banking (average of all time periods per segment)

	Brand awareness	Mental market share	Associative penetration	Associative rate	Purchase propensity
	-	All 0.83**	All 0.84**	All 0.79**	All 0.84**
Brand awareness		NU 0.42	NU 0.57	NU 0.36	NU -
		LU 0.48	LU 0.21	LU 0.23	LU 0.35
		HU 0.92**	HU 0.32	HU 0.01	HU 0.92**
	-		All 0.99**	All 0.97**	All 0.95**
Mental market share			NU 0.94**	NU 0.90**	NU -
			LU 0.76**	LU 0.21	LU 0.92**
			HU 0.25	HU 0.10	HU 0.98**
	-			All 0.96**	All 0.94**
Associative penetration				NU 0.84**	NU -
				LU 0.35	LU 0.71**
				HU 0.13	HU 0.26
	-				All 0.92**
Associative rate					NU -
					LU 0.33
					HU 0.06
Purchase propensity					-

All = whole sample; NU = non-users; LU = light users; HU = heavy users

** = correlation is significant at the 0.01 (two-tailed); * = correlation is significant at the 0.05 level (two-tailed)

Table 6 – Outcome of the regression analysis for soft drinks

		Brand awareness			Mental market share			Associative penetration			Associative rate			Model fit
		St.	Unst.	t-value	St.	Unst.	t-value	St.	Unst.	t-value	St.	Unst.	t-value	
Time 1	SD All	0.225	0.74	1.434	0.717	1.205	1.011	-0.585	-0.128	-1.051	0.544	0.055	1.779	0.789**
	SD Light	-0.126	-0.024	-1.531	1.217	0.904	11.608**	0.420	0.007	0.612	-0.328	0.017	-4.458**	0.951**
	SD Heavy	0.157	0.036	4.160	0.918	0.738	19.161**	-0.019	-0.005	-0.272	-0.110	0.004	3.580**	0.985**
Time 2	SD All	0.382	0.112	2.195	-0.643	-0.89	-0.864	0.579	0.120	1.081	0.769	0.079	2.827	0.821**
	SD Light	-0.075	-0.014	-0.403	1.051	0.806	4.052**	-0.381	-0.070	-2.748*	0.176	0.007	1.014	0.845**
	SD Heavy	-0.009	-0.002	-1.131	1.050	0.744	13.873**	-0.120	-0.028	-2.149*	-0.133	-0.004	-2.694	0.964**
Time 3	SD All	-0.185	-0.048	-0.792	1.612	2.175	1.899	-1.221	-0.258	-1.998	0.429	0.037	1.519	0.752**
	SD Light	-0.290	-0.057	-2.757*	1.373	0.980	11.322**	-0.127	-0.021	-2.27*	-0.227	-0.011	-3.044**	0.949**
	SD Heavy	-0.100	-0.020	-1.320	1.157	0.866	13.052**	-0.155	-0.300	-2.487*	-0.245	-0.009	1.989	0.951**

Note: SD = soft drinks; All = whole sample; Light = light users; Heavy = heavy users

Note: St. = standardized parameter estimate; Unst. = unstandardized parameter estimate; ** = $p < 0.001$

Table 7 – Outcome of the regression analysis for banking

		Brand awareness			Mental market share			Associative penetration			Associative rate			Model fit
		St.	Unst.	t-value	St.	Unst.	t-value	St.	Unst.	t-value	St.	Unst.	t-value	
Time 1	Bank All	0.095	0.1	0.437	-0.033	-0.054	-0.025	0.835	0.501	0.738	0.058	0.009	0.11	0.881**
	Bank Light	0.076	-0.042	-0.019	0.952	0.755	10.229**	-0.084	-0.027	-0.881	-0.274	0.025	3.626**	0.958**
	Bank Heavy	-0.045	-0.014	-0.202	1.015	0.994	4.682**	0.001	0.000	0.087	-0.037	-0.003	-0.464	0.960**
Time 2	Bank All	0.041	0.04	2.7	1.535	2.355	2.502*	-0.081	-0.051	-0.18	-0.532	-0.7	-1.365	0.953**
	Bank Light	0.040	0.017	0.779	0.438	0.231	4.209**	-0.550	-0.079	-5.454*	-0.143	-0.009	-2.783*	0.983**
	Bank Heavy	0.081	0.020	0.649	8.970	0.821	7.149*	0.112	0.053	0.859	-0.136	-0.006	-1.154	0.977**
Time 3	Bank All	0.207	0.171	1.358	1.656	2.324	1.798	-1.001	-0.563	-1.208	0.126	0.015	0.382	0.936**
	Bank Light	-0.061	-0.029	-0.412	0.927	0.376	4.326**	-0.018	-0.004	0.090	0.850	0.004	0.489	0.844**
	Bank Heavy	0.200	0.005	0.086	0.933	0.919	3.861**	-0.103	-0.043	-1.351	-0.067	-0.006	-0.918	0.971**

Note: Bank = banking; All = whole sample; Light = light users; Heavy = heavy users

Note: St. = standardized parameter estimate; Unst. = unstandardized parameter estimate; ** = $p < 0.001$